

SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I _S.VENKATESWARLU (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:S.VENKATESWARLU

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Question 1

Problem Statement

A neural network predicts student scores with a **Mean Squared Error (MSE)** of **0.85 during training** and **5.20 during testing**. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Solution

Introduction

A Neural Network is a Machine Learning model that learns patterns from training data and uses those patterns to predict outcomes for new data. The performance of a neural network is commonly measured using **Mean Squared Error (MSE)**, which calculates the average of the squared differences between the actual values and the predicted values.

Formula of Mean Squared Error (MSE)

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- n = Number of observations
- y_i = Actual value
- $\hat{y}_i$ = Predicted value

A **lower MSE** indicates better prediction accuracy, while a **higher MSE** indicates poor prediction performance.

Given

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

Problem Understanding

The neural network achieves an MSE of **0.85** on the training data, which means it predicts the training samples very accurately. However, the testing MSE is **5.20**, which is significantly higher. This large difference indicates that the model performs well only on the training data but fails to make accurate predictions on new or unseen data.

This situation is a classic example of **Overfitting**.

Error Identification

The following observations can be made:

- Training error is **very low**.
- Testing error is **much higher**.
- The model does not generalize well to unseen data.
- The neural network has learned the training data too closely instead of learning the underlying patterns.

Therefore, the primary error is **Overfitting**.

Error Analysis and Reasoning

Several factors may have caused the high testing error:

1. Overfitting

The model memorizes the training dataset rather than learning general relationships between input features and output values. As a result, it performs poorly when new data is presented.

2. Small Training Dataset

If the dataset contains only a limited number of student records, the neural network cannot learn sufficient patterns. A larger dataset generally improves the model's ability to generalize.

3. Complex Neural Network Architecture

Using too many hidden layers or neurons increases the complexity of the model. A highly complex model may fit every detail of the training data, including noise, resulting in poor testing performance.

4. No Regularization

Regularization techniques such as **Dropout** and **L2 Regularization** prevent the model from relying too heavily on specific neurons or weights. Without these techniques, overfitting becomes more likely.

5. Poor Quality Data

The training dataset may contain:

- Missing values
- Duplicate records
- Incorrect data
- Outliers
- Noise

These issues reduce the model's ability to make accurate predictions.

6. Improper Data Splitting

If the training and testing datasets do not represent the same distribution, the model may perform well during training but poorly during testing.

7. Excessive Training (Too Many Epochs)

Training the neural network for too many epochs causes it to memorize the training data instead of learning useful patterns. This increases the testing error.

8. Feature Selection Issue

Some input features may not be useful for predicting student scores. Including irrelevant features increases model complexity and reduces prediction accuracy.

Correction Strategy

To improve the neural network's performance, the following corrective measures are recommended:

1. Increase Training Data

Collect more student records so that the neural network learns a wider variety of patterns.

2. Apply Dropout

Dropout randomly disables neurons during training, forcing the network to learn more generalized features instead of memorizing the data.

3. Use L2 Regularization

L2 Regularization penalizes very large weight values and helps reduce overfitting.

4. Early Stopping

Monitor the validation error during training and stop training when the validation error begins to increase.

5. Simplify the Model

Reduce the number of hidden layers and neurons if the network is unnecessarily complex.

6. Improve Data Quality

Perform proper data preprocessing by:

- Handling missing values
- Removing duplicate records
- Eliminating outliers
- Normalizing data
- Cleaning noisy data

7. Perform Cross-Validation

Use techniques such as **K-Fold Cross Validation** to evaluate the model on different subsets of the dataset and obtain a more reliable estimate of performance.

8. Feature Engineering

Select only the most relevant features that influence student performance, such as attendance, previous marks, study hours, and assignment scores.

9. Hyperparameter Tuning

Optimize important parameters such as:

- Learning rate
- Batch size
- Number of epochs
- Number of hidden neurons
- Activation functions

This helps improve both training and testing performance.

Interpretation of Results

The large difference between training MSE (**0.85**) and testing MSE (**5.20**) clearly indicates that the neural network has not generalized well to unseen data. Although the training accuracy is high, the model cannot make reliable predictions for new student records.

By applying regularization methods, improving data preprocessing, increasing the dataset size, simplifying the model architecture, and tuning hyperparameters, the testing error can be significantly reduced. A good neural network should have **both low training error and low testing error**, with only a small difference between them.

Advantages of the Suggested Corrections

- Improves prediction accuracy.
- Reduces overfitting.
- Enhances model generalization.
- Produces reliable predictions on unseen student data.
- Increases robustness and stability of the neural network.
- Improves overall model performance.

Conclusion

The neural network achieved a **Training MSE of 0.85** but a **Testing MSE of 5.20**, indicating that it suffers from **overfitting**. The model has learned the training data very well but cannot generalize to new data. The main reasons include overfitting, insufficient training data, excessive model complexity, poor data quality, lack of regularization, and improper training strategies. By implementing techniques such as **Dropout, L2 Regularization, Early Stopping, Cross-Validation, Hyperparameter Tuning, Feature Engineering, and better data**

preprocessing, the gap between training and testing errors can be minimized, resulting in a more accurate and reliable neural network model.